

30 Machine Learning Mini-Capstone Project Topics (L1)

Regression Projects

1. Simple House Price Prediction: Given a small dataset with a few key features like size, number of bedrooms, and location (categorical), build a Simple or Multiple Linear Regression model to predict house prices. Focus on basic data cleaning and model training.
2. Predicting Student Exam Scores: Based on study hours, previous grades, and attendance, build a Linear Regression model to predict a student's final exam score.
3. Basic Car Mileage Prediction: Using car features like engine size, weight, and year, predict its Miles Per Gallon (MPG) using a Linear Regression model.
4. Sales Prediction for a Small Shop: Given historical daily sales data and day of the week, predict tomorrow's sales using Linear Regression.
5. Predicting Pizza Delivery Time: Based on order size, distance, and time of day, build a simple regression model to estimate pizza delivery time.
6. Predicting Body Fat Percentage: Using simple measurements like height, weight, and waist circumference, build a linear regression model to predict body fat percentage.
7. Estimating Restaurant Tip Amount: Based on bill total, number of diners, and day of the week, predict the tip amount using a simple regression model.

Classification Projects

8. Iris Flower Classification: A classic beginner projects. Given sepal and petal measurements, classify Iris flowers into one of three species using Logistic Regression, K-NN, or a simple Decision Tree.
9. Predicting Customer Subscription (Binary): Based on basic demographic information (age, income) and website activity (pages visited), predict if a customer will subscribe to a service (Yes/No) using Logistic Regression.
10. Titanic Survival Prediction: Using passenger data (age, gender, class), predict whether a passenger survived the Titanic disaster. Focus on basic feature engineering (e.g., handling missing age values) and a simple classification model.
11. Basic Email Spam Detection: Given a small set of emails (pre-processed into features like word counts), classify them as spam or not spam using Naive Bayes or Logistic Regression.
12. Predicting Loan Eligibility: Based on a few applicant features (e.g., income, credit score, employment status), classify if a loan application should be approved or rejected.
13. Wine Type Classification: Given chemical analysis data of different wines, classify them into their respective types (e.g., red/white, or specific varieties) using K-NN or Decision Tree.
14. Predicting Admission to University: Based on GPA, GRE scores, and university ranking, predict a student's chance of admission (admitted/not admitted) using Logistic Regression.
15. Breast Cancer Prediction (Simplified): Using a simplified version of the Wisconsin Breast Cancer dataset (fewer features), classify a tumor as benign or malignant using Logistic Regression or SVM.

Clustering & Dimensionality Reduction Projects

16. Basic Customer Grouping: Given a small dataset of customer purchase frequency and average spending, apply K-Means clustering to identify 2-3 distinct customer groups.
17. Grouping Similar Songs (Simple Features): Using very basic song features (e.g., duration, beats per minute), apply K-Means to group songs into a few clusters.
18. Simple Image Color Quantization: Take a small image and apply K-Means clustering to its RGB pixel values to reduce the number of colors in the image, demonstrating basic color compression.
19. Identifying Similar News Articles (Keywords): Given a small collection of news article headlines, extract keywords and use K-Means to group articles with similar topics.
20. Student Performance Grouping: Based on scores in different subjects, use K-Means clustering to group students into performance categories (e.g., high, medium, low).
21. Visualizing High-Dimensional Data (PCA on Iris): Apply PCA to the Iris dataset to reduce its 4 features to 2, and then visualize the data points in 2D space, observing how the species separate.
22. Simplifying a Dataset with PCA: Choose a dataset with more than 5-6 numerical features (e.g., a simple medical dataset) and apply PCA to reduce it to 2-3 principal components, explaining the variance retained.

Neural Networks Basics & Mixed Projects

23. AND/OR/XOR Gate with Perceptron (Manual/Simple Code): Implement a simple Perceptron from scratch (or using NumPy) to learn the logic of AND/OR gates. Discuss why XOR is not linearly separable.
24. Basic Digit Classification (MNIST - Simplified with MLP): Use the MNIST handwritten digit dataset. Build a very simple Multi-Layer Perceptron (MLP) with one hidden layer using TensorFlow/Keras to classify digits. Focus on the setup and training process.
25. Predicting Diabetes Onset (Pima Indians Dataset): Given the Pima Indians Diabetes dataset, build a simple classification model (e.g., Logistic Regression or a small MLP) to predict if a patient has diabetes.
26. Simple Image Classification (CIFAR-10 - Binary Subset): Take a subset of the CIFAR-10 dataset (e.g., only classify "cat" vs. "dog" images) and train a very basic CNN with 1-2 convolutional layers using TensorFlow/Keras.
27. Predicting Heart Disease (Simplified): Using a simplified heart disease dataset, build a classification model (e.g., Logistic Regression or Decision Tree) to predict the presence of heart disease.
28. Predicting Abalone Age (Regression): Given physical measurements of abalone, predict their age (number of rings) using a simple regression model like Linear Regression or a small MLP.
29. Text Sentiment Analysis (Positive/Negative - Simple Dataset): Given a small, pre-processed dataset of positive and negative movie reviews, build a simple classification model (e.g., Logistic Regression or MLP) to classify sentiment.
30. Basic Recommendation System (Content-Based): Given a small dataset of movies and their genres,

recommend movies to a user based on the genres of movies they have liked previously. (Focus on simple similarity calculation).

30 Machine Learning Mini-Capstone Project Topics (L2)

Regression Projects

1. **Predicting House Prices with Advanced Features:** Given a dataset with raw house features (e.g., King County House Sales or Ames Housing), perform extensive feature engineering (e.g., creating age of house, total square footage, neighborhood encoding), handle outliers, and build a robust regression model (e.g., Gradient Boosting Regressor or XGBoost) to predict sale prices. Evaluate with RMSE and R-squared.
2. **Forecasting Energy Consumption:** Using historical energy usage data and external factors (temperature, day of week, holidays), build a time-series regression model (e.g., Linear Regression with time-based features, or a simple MLP) to predict future energy consumption for a building or city block.
3. **Predicting Customer Lifetime Value (CLTV):** Based on historical transaction data (purchase frequency, average order value, recency), develop a regression model to estimate the future revenue a customer will generate over a specific period.
4. **Estimating Used Car Prices:** Given a dataset of used car listings with various attributes (make, model, year, mileage, engine size, features), build a regression model to predict the selling price. Focus on handling categorical features and potential non-linear relationships.
5. **Predicting Stock Market Volatility (Simplified):** Using historical stock prices (Open, High, Low, Close, Volume) for a specific stock, build a regression model to predict its daily price range (High - Low) or closing price based on previous days' data. (Note: This is a simplified approach, not for actual trading).
6. **Predicting Restaurant Sales:** Given historical sales data, promotional events, weather conditions, and day of the week, build a regression model to predict daily or weekly sales for a restaurant.
7. **Movie Revenue Prediction:** Based on movie attributes like genre, director, cast, budget, release date, and critical reception scores, predict a movie's box office revenue.

Classification Projects

8. **Credit Card Fraud Detection:** Given a highly imbalanced dataset of credit card transactions, build a classification model (e.g., Logistic Regression, Random Forest, or SVM) to identify fraudulent transactions. Focus on handling class imbalance (e.g., SMOTE, undersampling) and evaluating with Precision, Recall, and F1-score.
9. **Customer Churn Prediction:** Using customer demographic data, service usage patterns, and historical churn information, build a classification model to predict which customers are likely to churn. Emphasize feature importance and actionable insights.
10. **Spam Email Classification:** Given a dataset of emails labeled as spam or ham, preprocess the text data (tokenization, TF-IDF or CountVectorizer) and build a classification model (e.g., Naive Bayes, SVM, or Logistic Regression) to categorize new emails.
11. **Diabetic Retinopathy Detection (Simplified Image Classification):** Given a dataset of retinal images labeled with different stages of diabetic retinopathy (or just healthy/diseased), build a simple Convolutional Neural Network (CNN) using TensorFlow/Keras to classify the images. (Focus on

basic CNN architecture, not complex models).

12. **Sentiment Analysis of Movie Reviews:** Given a dataset of movie reviews and their corresponding sentiment (positive/negative), preprocess the text and build a classification model (e.g., Logistic Regression, SVM, or a simple MLP) to predict the sentiment of new reviews.
13. **Loan Default Prediction:** Using applicant financial information, credit history, and loan details, build a classification model to predict the likelihood of a loan applicant defaulting.
14. **Online News Popularity Prediction:** Based on features like number of words, number of images, categories, and social media shares, predict whether an online news article will be highly popular (e.g., top 10% of shares).
15. **Malware Detection:** Given a dataset of software features (e.g., API calls, permissions) labeled as benign or malicious, build a classification model to identify malware.

Clustering & Dimensionality Reduction Projects

16. **Customer Segmentation:** Using customer demographic and transaction data (e.g., RFM - Recency, Frequency, Monetary value), apply K-Means clustering to identify distinct customer segments. Analyze and characterize each segment based on their attributes.
17. **Document Clustering:** Given a collection of text documents (e.g., news articles, research papers), preprocess the text (TF-IDF) and apply K-Means or Hierarchical Clustering to group similar documents together. Visualize the clusters using PCA or t-SNE.
18. **Image Compression using K-Means:** Apply K-Means clustering to the pixel colors of an image to reduce the number of unique colors, effectively compressing the image while retaining visual quality. (This involves treating each pixel's RGB values as a data point).
19. **Anomaly Detection in Network Traffic:** Using network traffic data (e.g., packet size, protocol, source/destination IP), apply clustering algorithms (e.g., K-Means or DBSCAN) to identify unusual patterns that might indicate network intrusion or anomalies.
20. **Gene Expression Data Analysis:** Given a dataset of gene expression levels across different samples (e.g., healthy vs. diseased), apply PCA to reduce dimensionality and then K-Means clustering to identify groups of similar samples or genes.
21. **Crime Hotspot Identification:** Using geographical crime data (latitude, longitude, crime type), apply clustering algorithms to identify areas with high crime density (hotspots).
22. **Music Genre Clustering:** Given a dataset of songs with audio features (e.g., tempo, energy, danceability), apply clustering to group songs into distinct genres or moods.

Mixed/Advanced Projects (Combining Concepts – L3)

23. **Predicting Wine Quality:** Given chemical analysis of wine samples and their quality ratings, build a classification model (e.g., RandomForestClassifier) to predict wine quality categories (e.g., good, medium, bad) or a regression model to predict the numerical quality score. Use feature engineering and hyperparameter tuning.
24. **Handwritten Digit Recognition (MNIST):** Use the MNIST dataset of handwritten digits. Apply PCA for dimensionality reduction and then train a classification model (e.g., SVM, Logistic Regression, or a simple MLP in TensorFlow/Keras) to classify the digits. Compare performance with and without PCA.
25. **Predicting Forest Cover Type:** Given geographical and environmental features of forest areas, build a multi-class classification model to predict the forest cover type. This involves handling a multi-class

problem and potentially many features.

- 26. Recommendation System (User-Based or Item-Based Collaborative Filtering - Simplified):** Given a dataset of user ratings for items (e.g., movies, products), implement a simplified version of user-based or item-based collaborative filtering to recommend items to a user. (Focus on the core logic, not a full-scale system).
- 27. Predicting Customer Satisfaction:** Using survey responses and customer interaction data, build a classification model to predict customer satisfaction levels (e.g., satisfied, neutral, dissatisfied).
- 28. Medical Diagnosis Prediction:** Given patient symptoms, lab results, and medical history, build a classification model to predict the presence of a specific disease (e.g., heart disease, breast cancer). Focus on interpretability and appropriate evaluation metrics for medical contexts.
- 29. Traffic Accident Severity Prediction:** Using accident data (weather, road conditions, time of day, location), build a classification model to predict the severity of a traffic accident (e.g., minor, moderate, severe).
- 30. Text-Based Job Role Classification:** Given job descriptions, preprocess the text and classify them into predefined job roles or industries using a multi-class classification model.